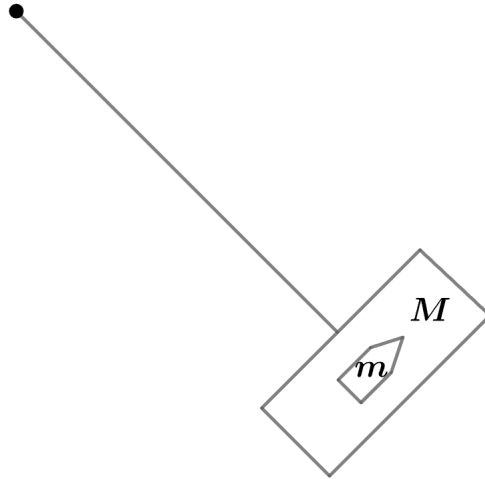


2009 F=ma Exam: Problem 14

Kevin S. Huang



- A) When the block-bullet system reaches its peak height, the momentum is zero. Since the bullet initially had momentum, the total linear momentum is not conserved.
- B) Horizontal component of linear momentum not conserved because of same reason as A).
- C) Since the bullet embeds in the block, we have a perfectly inelastic collision which does not conserve mechanical energy.
- D) When the block-bullet system reaches its peak height, the angular momentum is zero. Since the bullet initially had angular momentum about the peg, the total angular momentum is not conserved.
- E) This is true since none of the above quantities are conserved.

Thus, the answer is E.